

AICA LEVEL 2 CAPSTONE PROJECT

AI-Powered Invoice Extras Extraction

Hybrid Rule Engine + Generative AI + Validation + Human Review

Project Type	AICA Level 2 Capstone
Domain	Accounting / Logistics Automation
Prototype	Desktop Python application
Test Outcome	8 of 8 defined scenarios passed

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Prepared as a demonstration prototype for controlled extraction of customer additional charges from unstructured logistics emails.

1. Executive Summary

This capstone demonstrates a controlled AI-assisted workflow for extracting invoice-extra information from logistics emails. The prototype identifies file references, container numbers and additional customer charges, calculates the total, evaluates uncertainty, applies deterministic validation rules, and routes exceptions to a human reviewer for approval or rejection.

The design deliberately separates extraction from control. A rule-based engine handles predictable formats, while a Gemini-based AI extractor interprets less structured language. A separate validation module checks mandatory fields, confidence thresholds, zero or negative values and duplicate charges. The AI cannot override these deterministic controls.

Eight defined test scenarios were executed through the AI and validation pipeline. All eight produced the expected total and the expected review decision. This 8/8 result applies only to the defined test pack and must not be interpreted as universal 100% accuracy on real-world data.

2. Business Problem

In transport and clearing operations, additional customer-recoverable charges may arrive through email in different formats. Examples include ferry charges, demurrage, storage, detention, warehouse rent and port penalties. Manual review is repetitive and creates a risk of omission, duplicate recovery, incorrect amounts, or acceptance of uncertain information.

- Emails may be structured, semi-structured or written in informal natural language.
- Important identifiers such as file reference or container number may be missing.
- Quoted email history may contain older amounts that should not be treated as current charges.
- Zero-value items should not be included in invoice extras.
- Approximate or ambiguous amounts should be reviewed rather than automatically accepted.
- Duplicate charges should be identified before downstream use.

3. Project Objectives

- Extract file reference, container number, charge description and USD amount from email text.
- Use deterministic rules for predictable data and generative AI for unstructured language.
- Exclude zero-value charges.
- Calculate total customer extras from positive extracted charges.
- Apply independent validation controls after AI extraction.
- Route exceptions to a human reviewer.
- Provide explicit Approve / Reject controls so AI is advisory rather than autonomous.
- Create repeatable test evidence for the capstone demonstration.

4. Solution Architecture

EMAIL INPUT -> RULE ENGINE / AI EXTRACTOR -> VALIDATION ENGINE -> APPROVED or NEEDS REVIEW -> HUMAN APPROVE / REJECT

4.1 Main Components

Module	Purpose	Control Value
rule_extractor.py	Pattern-based extraction of predictable email formats.	Fast, deterministic baseline.
ai_extractor.py	Gemini structured extraction for natural-language emails.	Interprets messy wording and returns confidence.
validation.py	Independent deterministic checks.	Prevents AI from self-approving exceptions.
app.py	Desktop GUI and human decision workflow.	Makes review status and final decision visible.
test_cases.py	Defined capstone scenarios.	Provides repeatable test inputs and expected outcomes.
run_tests.py	Automated AI/validation test runner.	Produces measurable PASS/FAIL evidence.

5. AI Extraction Design

The AI layer uses the Google GenAI Python SDK and requests structured JSON output mapped to Pydantic models. The prompt instructs the model to extract only customer additional charges, ignore zero amounts, prefer the latest message over quoted history, avoid double-counting written totals, standardize obvious spelling variations, and flag ambiguity for review.

The API key is read from the Windows GEMINI_API_KEY environment variable rather than being hard-coded in the source files. This reduces the risk of accidentally submitting or sharing a credential with the project code.

6. Validation and Human-in-the-Loop Controls

Control	Behavior
File reference validation	Missing or invalid file reference -> Needs Review.
Container validation	Missing or invalid container number -> Needs Review.
Zero/negative charge control	Amounts ≤ 0 are excluded or blocked.
Confidence threshold	Charge confidence below 85% -> Needs Review.
AI uncertainty flag	Approximate or ambiguous wording can trigger review.
Duplicate detection	Repeated description + amount -> Needs Review.
Human decision	Reviewer can Approve or Reject the AI result; validation issues require explicit confirmation before exception approval.

7. Demonstration Evidence

AI-Powered Invoice Extras Extraction

AI-Powered Invoice Extras Extraction

AICA Level 2 Capstone Project - Hybrid Rule + AI Extraction

Sample Email / Unstructured Input

Subject: Extra recovery SIM-0999-26 MSCU1234567

pls recover from client around 450 dollars being storage.
also there is 225 usd port penalty which client has to bear.

Rule Extraction

AI Extraction

Needs Review

Clear Results

Structured Extraction Result

Field	Extracted Value	Confidence	Status
File Reference	SIM-0999-26	85%	AI Confirmed
Container	MSCU1234567	85%	AI Confirmed
STORAGE	USD 450.00	80%	AI Extracted
PORT PENALTY	USD 225.00	95%	AI Extracted
Total Extras	USD 675.00	100%	Calculated
Review Reason	Storage charge amount is approximate ('around 450 dollars')	85%	Needs Review

Figure 1. AI extraction of an approximate storage amount. The system extracted the values but routed the case to Needs Review because the source used the word 'around'.

8. Test Methodology and Results

The automated test runner processed eight predefined scenarios through the AI extraction and validation pipeline. Each case compared the expected total and expected review decision with the actual outputs.

ID	Scenario	Expected USD	Actual USD	Exp. Review	Actual Review	Result
TC01	Standard Structured Email	2,106.00	2,106.00	No	No	PASS
TC02	Messy Natural Language	675.00	675.00	No	No	PASS
TC03	Approximate Amount	675.00	675.00	Yes	Yes	PASS
TC04	Zero Amount	800.00	800.00	No	No	PASS
TC05	Missing	350.00	350.00	Yes	Yes	PASS

	Container					
TC06	Missing File Reference	700.00	700.00	Yes	Yes	PASS
TC07	Possible Duplicate Charge	1,100.00	1,100.00	Yes	Yes	PASS
TC08	Multiple Charge Types	2,525.00	2,525.00	No	No	PASS

Test summary: 8 tests executed, 8 passed, 0 failed, 0 errors.

8.1 Key Exception Tests

- TC03: 'around 450 dollars' was extracted as USD 450, but uncertainty and low confidence correctly caused Needs Review.
- TC04: a USD 0 storage item was ignored while positive charges were retained.
- TC05: the correct USD 350 charge was extracted, but the missing container triggered a HIGH validation issue.
- TC06: the correct USD 700 charge was extracted, but the missing file reference triggered a HIGH validation issue.
- TC07: two identical USD 300 storage charges were retained in the total and the deterministic duplicate control flagged the case for review.

9. Accounting and Governance Relevance

The project is designed as decision support rather than autonomous accounting. It demonstrates segregation between extraction, validation and final human decision. This is important where AI-generated information may affect customer billing, recoverable expenses or management reporting.

- Traceable control logic: deterministic validation rules are separate from the generative model.
- Exception-based review: uncertain or incomplete records are not silently treated as clean data.
- Human accountability: the final Approve / Reject decision remains with the reviewer.
- No automatic posting to accounting records is included in this prototype.
- The prototype can be extended with audit logging, reviewer identity, timestamps and controlled export.

10. Security and Privacy

- The Gemini API key is stored outside source code in an environment variable.
- API keys, OAuth tokens and credentials should never be included in the submission ZIP.
- Real customer emails should be anonymized before inclusion in academic evidence.
- The prototype should use least-privilege access if later connected to live Gmail.
- Production deployment should define retention, access control, logging and data-processing requirements before live use.

11. Limitations

- Eight test cases are a useful prototype test pack but are not sufficient to establish production-level accuracy.
- Generative AI output and confidence can vary; deterministic validation and human review remain necessary.
- The current duplicate rule is intentionally simple and may need business-context logic for legitimate repeated charges.
- The prototype assumes USD for the demonstrated extraction workflow.
- The current application is a desktop prototype and does not yet include multi-user roles, centralized database storage or a complete audit trail.
- Model availability and API behavior may change over time, requiring maintenance.

12. Benefits Demonstrated

- Reduces repetitive reading of logistics emails.
- Converts unstructured messages into structured billing data.
- Combines deterministic automation with AI language understanding.
- Makes uncertainty visible rather than hiding it.
- Supports exception-based review and human accountability.
- Creates a foundation for future integration with costing, invoicing and reporting systems.

13. Future Enhancements

- Connect the prototype to approved Gmail sources using read-only or least-privilege access.
- Add a persistent audit log recording source message ID, extraction method, validation issues, reviewer decision and timestamp.
- Export approved records to Excel/database tables for downstream costing and invoice preparation.
- Add customer, destination, truck/trip and ownership master-data integration where appropriate.
- Introduce role-based access and multi-user review queues.
- Expand testing with anonymized historical emails and measure field-level precision, recall and exception-routing performance.

14. Conclusion

The capstone demonstrates a practical hybrid approach to AI-enabled accounting workflow automation. Rule-based extraction provides deterministic handling of predictable formats, generative AI interprets unstructured language, deterministic validation checks the result, and a human reviewer retains final control. The defined eight-scenario test pack passed in full, including deliberate exception cases. The prototype therefore provides a credible proof of concept while clearly preserving the controls required before any production use.

Appendix A - Project Files

- app.py - GUI and human review workflow
- rule_extractor.py - deterministic extraction
- ai_extractor.py - Gemini structured extraction
- validation.py - validation engine
- test_cases.py - eight defined test scenarios
- run_tests.py - automated test runner
- AICA_Level2_Capstone_Test_Evidence.xlsx - test evidence workbook